

Two-Dimensional Panoramic Image Stitching using PCA-SIFT Approach

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Abstract— Image stitching is a prominent field within computer vision and graphics, widely integrated into tools like cell phones and image processing APIs. This paper delves into image stitching using Python and OpenCV, aiming to merge images with shared regions into panoramic scenes. The technique merges photographs from different cameras to create seamless panoramas. The approach relies on invariant features, specifically the PCA-SIFT algorithm, which uses Scale Invariant Feature Transform (SIFT) for detecting and describing local image features. SIFT identifies significant points, extracting position, scale, and rotation invariants for feature descriptors. This method benefits both general image stitching and stitching based on local features. The Python-implemented PCA-SIFT outperforms standard SIFT in key point matching accuracy, leading to better retrieval results. Additionally, PCA-SIFT demonstrates faster matching performance. Here, the research objective is to enhance image stitching quality by using the PCA-SIFT algorithm, addressing the limitations of standard SIFT. The projected achievements include extracting key points using the SIFT algorithm, matching them with Euclidean distance, and comparing stitching quality between SIFT and PCA-SIFT.

Keywords— Image Stitching, OpenCV, Principal Component Analysis (PCA), Scale Invariant Feature Transform (SIFT), extreme points.

I. INTRODUCTION

Image stitching is the process of merging numerous photographs that possess overlapping fields of view, ultimately generating panoramas or high-resolution images. Commonly referred to as image mosaicking, this process typically necessitates precise image overlaps and identical exposures for seamless outcomes. Image stitching finds applications in computer vision, graphics, and some digital cameras, which can internally stitch their photos. While human vision covers around 135 x 200 degrees, most cameras offer only 35 x 50 degrees. Panoramic mosaicking involves capturing numerous images with an ordinary camera and combining them to achieve a broader field of view. Stitching quality is evaluated based on image similarity and seam visibility.

Medical image stitching plays a vital role in the field of clinical diagnostics, aiding healthcare professionals in assessing a wide range of conditions, including cardiac,

retinal, pelvic, renal, abdominal, liver, and tissue disorders. The continuous advancements in computer science have resulted in the creation of highly effective image processing methods, which are invaluable in supporting medical diagnoses, treatment planning, and facilitating research. The alignment of images for improved observation, known as image registration, is crucial. Alignment is essential even when comparing images taken at different times or from various devices and angles, including 2D or 3D perspectives.

However, image stitching faces challenges like noisy images, managing a large image database, high resolutions, parallax, scene motion, and eliminating visible seams. The aim is to achieve seamless panoramic images through automatic stitching using two input images based on feature extraction, particularly PCA-SIFT, to assess the method's effectiveness.

A. Feature Extraction

In general feature refers to information regarding the content of an image, often indicating specific attributes present within a particular region of the image [1]. These attributes can encompass structures like points, edges, or objects. Features might emerge as outcomes from operations applied to the image's local neighborhood or from feature detection methods. Feature detection, a crucial step in numerous computer vision algorithms, has led to the development of a wide array of feature detectors [13]. These detectors vary in their computational complexity, the types of features they identify, and their repeatability.

When features are derived through local operations on an image—commonly known as feature extraction, one can distinguish between feature detection approaches that produce local decisions whether there is a feature of a given type at a given image point or not, and those who produce non-binary data as result. The distinction becomes relevant when the resulting detected features are relatively sparse [2]. Although local decisions are made, the output from a feature detection step does not need to be a binary image. The result is often represented in terms of sets of (connected or unconnected) coordinates of the image points where features have been detected, sometimes with subpixel accuracy. When feature extraction is done without local decision making, the result is often referred to as a feature image.

Consequently, a feature image can be seen as an image in the sense that it is a function of the same spatial variables as the original image, but where the pixel values hold information about image features instead of intensity or color. This means that a feature image can be processed in a similar way as an ordinary image generated by an image sensor. Feature images are also often computed as integrated step in algorithms for feature detection. Feature extraction typically follows the pre-processing phase in character recognition systems. Pattern recognition, which involves assigning an input pattern to one of several potential output classes, is split into two key stages: feature selection and classification. Effective feature selection is pivotal since a poorly chosen set of features can hinder the classifier's recognition capabilities.

Various techniques for feature extraction exist, including PCA (Principal Component Analysis) and SIFT (Scale-Invariant Feature Transform). SIFT, in particular, finds extensive use in object recognition and other applications. PCA is notably efficient and accurate, making it a preferred choice for tasks like image stitching [16]. To sum up, PCA and SIFT techniques stand out due to their efficiency and accuracy, setting them apart in the realm of feature extraction [7] techniques.

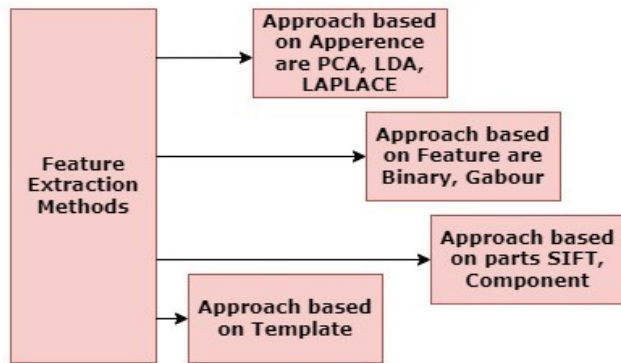


Fig1. Feature Extraction Techniques.

B. Image Stitching

Image stitching is a powerful technique that allows for the fusion of multiple individual shots into a single, expanded image, transcending the confines of typical aspect ratios and resolutions. Through the seamless fusion [11] of two or more images captured from the same location, each with overlapping fields of view, a high-resolution composite image is created. This composite, often referred to as a panorama or mosaic, captures a broader view and encompasses more visual information. This process enhances the overall visual experience and provides a panoramic [6] representation of the captured scene. As depicted in Fig. 2, the technique of image stitching brings together these individual shots into a cohesive and detailed final result.



Fig2. Example Image of Image Stitching.

C. Problem Identification

Standard SIFT usage for image stitching lacks precise matching accuracy at the key point level. This results in poor image stitching quality [9], as illustrated in Figure 2.1. The computational speed of SIFT is slow. Moreover, the high uniqueness and abundant information characteristics of SIFT aren't well-suited for rapid and accurate matching in extensive feature databases.



Fig3. Image of Stitched Image Using SIFT.

D. SIFT Algorithm:

SIFT is a powerful tool for image alignment and 2-D object recognition. Lowe introduced this method to extract features that remain invariant to changes in scale and rotation, exhibit robustness against occlusion and clutter, and operate efficiently in real-time applications. The algorithm constructs a Gaussian scale space, finds extreme points, eliminates unstable features, determines key point directions, and generates descriptors. These key points remain prominent and unaffected by various factors. SIFT's key points are used for image localization [10] and are invariant to image size and orientation. These features prove exceptionally valuable for computer vision applications such as image matching [4], object detection, and scene recognition. These key points can also be utilized during model training for feature extraction. SIFT's key points are advantageous as they remain unaffected by image size or orientation, achieved through scale-invariant coordinates. The SIFT Detector's design is inspired by the primate visual system. SIFT can detect objects [5][8] in clutter and partial occlusion, with its feature descriptor being invariant to scaling, orientation, illumination changes, and partially to

affine distortion., key point localization [14] in SIFT involves detecting distinctive and stable points in an image while accounting for variations in scale and orientation, making SIFT a powerful tool for various computer vision tasks.

Steps in SIFT Algorithm are given below as

1. Scale Space Peak Localization.
2. Key Point Localization.
3. Assigning Orientation to Key Points.
4. Key Point Descriptor.

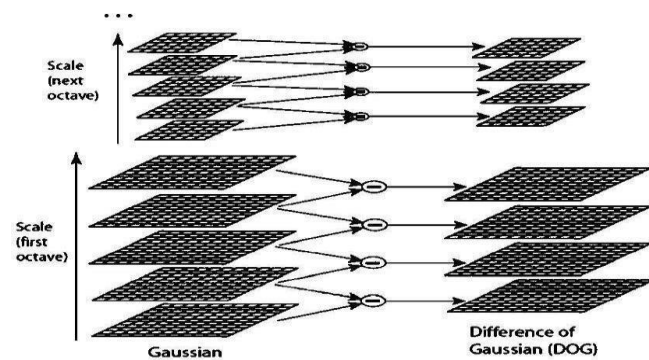


Fig4. Scale Space Peak Localization.



Fig5. Key Point Localization.

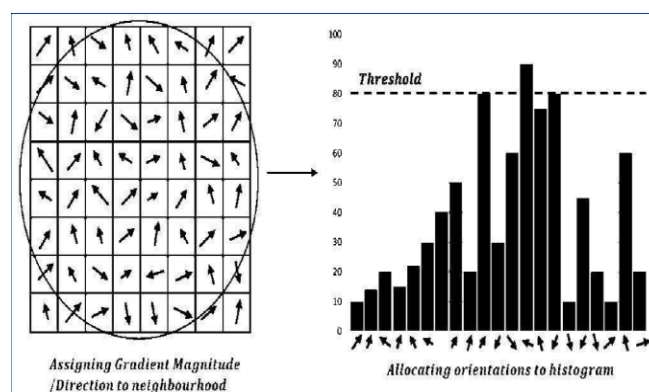


Fig6. Assigning Orientation to key points.

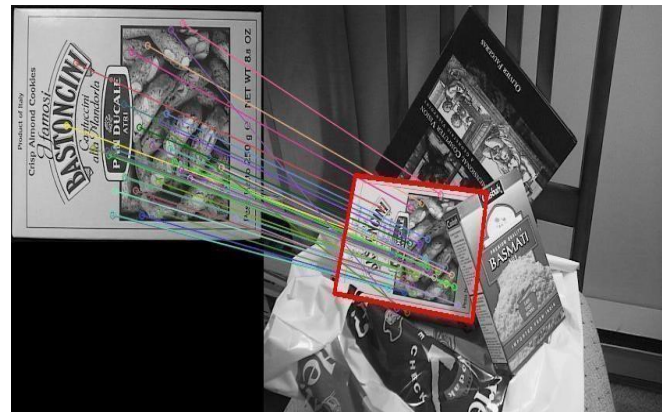


Fig7. Object Recognition Using SIFT.

E. PCA Algorithm

Principal Component Analysis (PCA) is a dimensionality reduction technique commonly applied to large datasets. It transforms a set of variables into a smaller one, retaining most of the information. While reducing variables sacrifices some accuracy, the aim is to simplify data. This enables easier exploration, visualization, and quicker analysis for machine learning algorithms. By eliminating extraneous variables, PCA streamlines the process. In essence, PCA's [15] goal is to reduce data variables while retaining maximal information.

Steps in PCA Algorithm are given below as

1. Standardization.
2. Computation of the Covariance Matrix.
3. Calculation of Eigen Vectors and Eigen Values from the Covariance Matrix to Determine Principal Components.
4. Creation of the Feature Vector.
5. Transformation of the Data Along the Principal Component Axes.

II. LITERATURE SURVEY

In 2004, D. G. Lowe introduced "Distinctive image features from scale invariant key points," presenting SIFT key points for accurate matching from an extensive database. These key points are distinct due to high-dimensional vectors representing local image gradients. They are invariant to rotation and scale, robust to distortion, noise, and illumination changes. Extracted from images, they work well for small objects in clutter and larger key points handle noise and blur. Computation is efficient, extracting thousands of key points in near real-time on standard PCs.

In 2013, Russol Abdelfatah and Haitham Omer published "Automatic Seamless Image Stitching," aiming to achieve high-quality image stitching through feature-based alignment and blending algorithms. The method involves extracting features and using Harris corner detectors for feature detection, followed by alignment of neighboring pairs using the RANSAC algorithm. Linear blending is then applied to eliminate transitions between aligned images. The algorithm successfully creates panorama images from captured images at a fixed linear spatial interval.

In 2015, Y. D. Feng introduced "Image Matching Algorithm Based on SURF Feature Extraction and FLANN Search" to address issues of limited feature information and high false match rates in traditional image matching. The algorithm utilizes SURF for feature extraction, involving extremum value computation from the Hessian matrix and processing feature vectors in multi-scale image space. The FLANN algorithm is employed for feature matching. Experimental results demonstrate improved effectiveness and efficiency compared to traditional methods.

Real-world applications demand increased robustness for correctly matching SIFT features. This paper introduces an enhancement to the original SIFT algorithm to ensure more dependable feature matching, [12] specifically for object recognition. The central concept involves partitioning features from both the test and model object images into distinct sub-collections prior to matching. These sub-collections are organized based on features stemming from various octaves, representing different frequency domains.

III. PROPOSED SYSTEM

To enhance the quality of image stitching [3] via the Scale Invariant Feature Transform (SIFT) algorithm, we propose integrating the Principal Component Analysis (PCA) algorithm within SIFT's Key-Point descriptor stage. This integration simplifies the descriptor representation, making it faster and more accurate compared to the standard SIFT descriptor. The use of PCA in the SIFT algorithm speeds up computations, leading to effective noise-free image stitching quality.

The incorporation of PCA into SIFT involves these steps:

1. Pre-compute an eigenspace for expressing local patch gradient images.
2. Compute the local image gradient for a given patch.
3. Project the gradient image vector or key point descriptor using the eigenspace to obtain a compact feature vector.

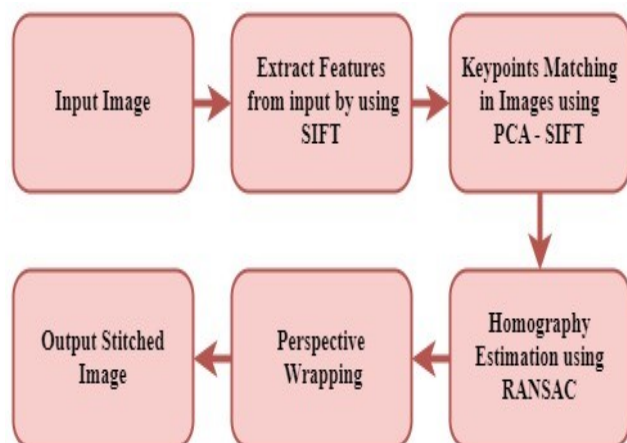


Fig8. Proposed Block Diagram.

The algorithm for the new method is outlined as follows:

1. Start
2. Input the given images.

3. Utilize the SIFT algorithm to identify unique features in the input images, known as Key-Points.
4. Display the detected Key-Points in the input images.
5. If there are a sufficient number of matching points in the input images, proceed to step 7.
6. If there are not enough matching points, adjust the position of input images with some overlap, and then go to step 10.
7. Apply the Homographic RANSAC algorithm to the detected Key-Points.
8. Implement Perspective Warping in the process.
9. Display the resulting stitched image.
10. End the process.

$$\text{Euclidean Distance} = \text{SQRT}(((x_2 - x_1)^2 + (y_2 - y_1)^2)).$$

Here (x1, y1) and (x2, y2) are two points.

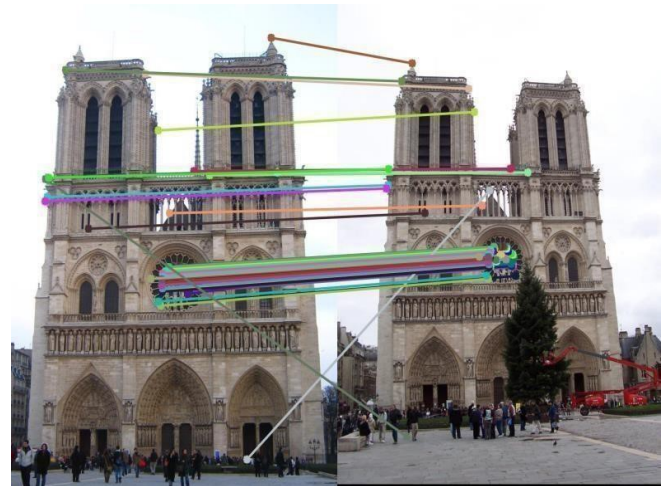


Fig9. Key Points Matching

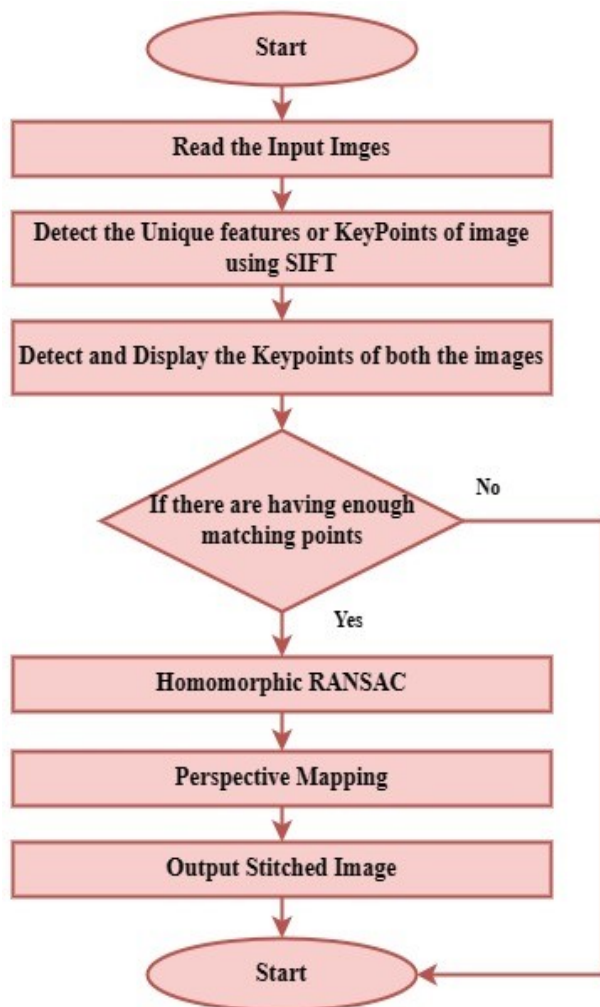


Fig10. Flowchart.

A. Software Requirements

In our paper, Visual Studio Code and OpenCV were employed as the tools for the 2-D implementation of panoramic image stitching.

To implement a PCA-SIFT-based 2D panoramic image stitching solution, VS Code is used for code composition and compilation, and involves following steps,

- Open VS code and create a new python project.
- Begin by importing the required libraries for image processing and feature extraction, including OpenCV.
- Implement PCA-SIFT for feature extraction and matching, involving the computation of principal components for SIFT features in each image and matching between image pairs.
- Implement the image blending algorithm to create the panoramic image. This involves aligning the images using the matched features and then blending them together to create a seamless panorama.
- Compile and run the code to test the implementation.

OpenCV, short for Open-Source Computer Vision Library, is a versatile open-source software library for computer vision and machine learning. It provides over 2500 optimized algorithms, covering classic and modern computer vision and machine learning tasks like face

detection, object identification, human action recognition, camera tracking, 3D modeling, image stitching, and more. OpenCV supports multiple programming languages such as Python, C++, and Java. When combined with libraries like NumPy for numerical operations, it enhances your capabilities for image and video processing, object recognition, and data manipulation.

IV. RESULTS AND DISCUSSION

To achieve a 2-D realization of panoramic image stitching, the initial step involves extracting significant key points from the input images. Subsequently, these key points are matched. Following this, homography estimation and perspective warping are applied to generate the ultimate stitched image output.

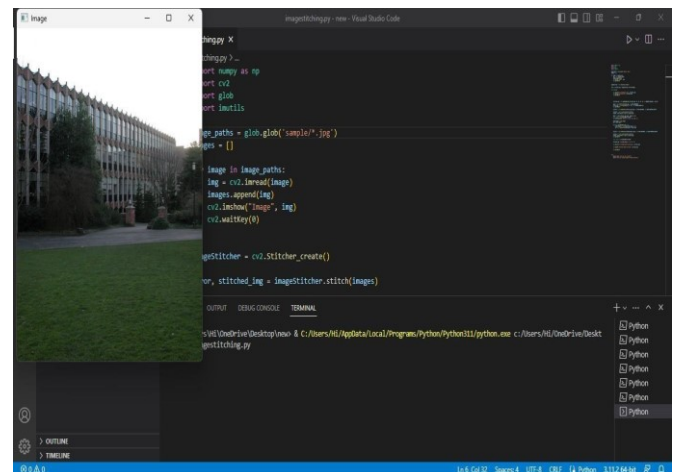


Fig11. Input Image 1.

The above image contains the unique key points which are nothing but unique patches in the image. This key – points will be generated by applying difference of gaussian to image at different sigma values, hence there will be stack of images will be formed and from which the peak point nothing but the local extreme will be found out and it is the desired key point. SIFT algorithm will help to find out this key-points in the image efficiently and accurately for matching purpose and stitching purpose.

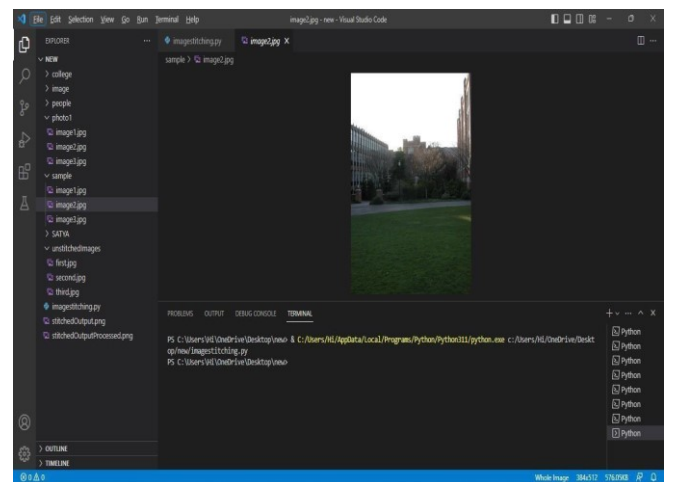


Fig12. Input Image 2.

The above image is second input image with key points which are nothing but the unique patches in the image two. The process used to extract key points in image one will

also be applied to image two for extracting key points. If there are no local extrema, then there is no unique feature at that pixel value. If there are not enough key points are generated, then it won't be further processed for matching and stitching purposes. After, extracting the enough key points an image then it is important to remove invalid key points in the image.

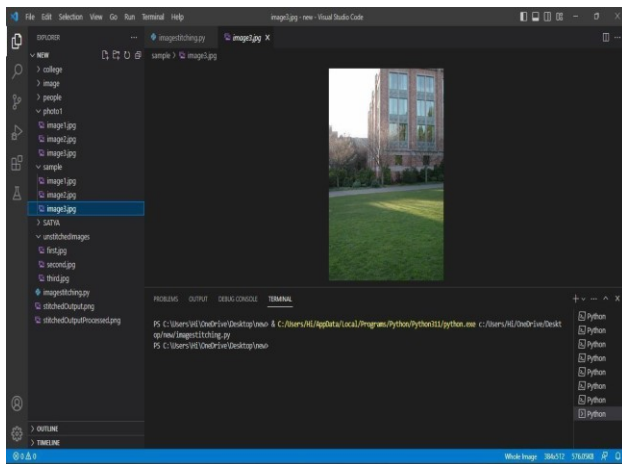


Fig13. Input Image 3.

The above figure is the third input image with key points which are nothing but the unique patches in the image three.

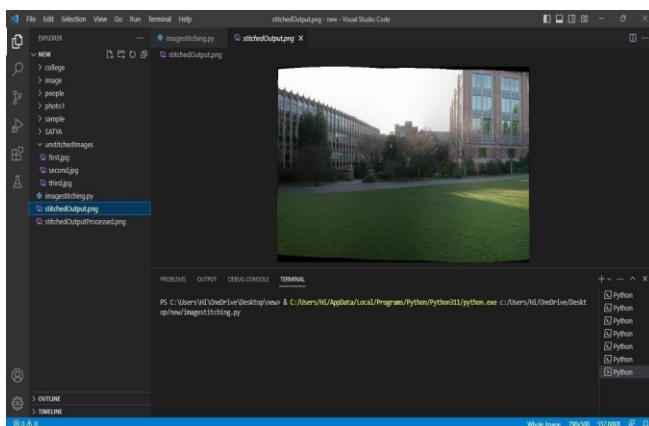


Fig14. Stitching Image.

The depicted figure demonstrates the stitched output image achieved through SIFT. Matching occurs among three input images, with histograms generated for key points across all images. These histograms are then compared using the Euclidean distance formula; if the output is high, corresponding key points are considered matched. A descriptor step incorporates Principal Component Analysis (PCA) to eliminate redundancy, thereby enhancing computational efficiency and matching accuracy.

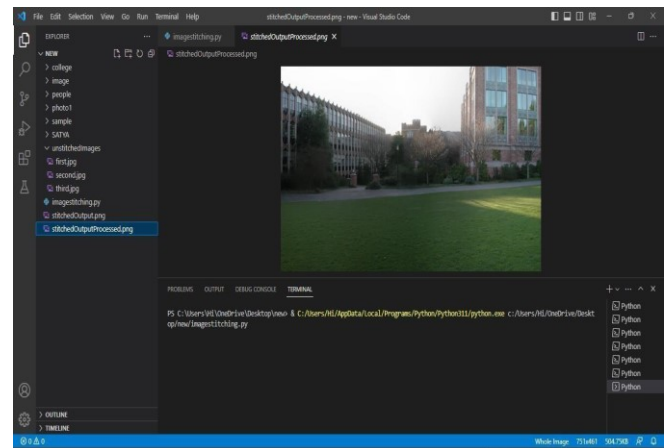


Fig15. Output Stitched Image.

The depicted diagram represents the resultant image, which is essentially the outcome of stitching together the provided input images. The application of the RANSAC algorithm serves to eliminate inaccurate matches in the images, retaining only the valid ones. By calculating Homography, the images are aligned precisely based on their coordinates. Following this alignment, the images are seamlessly stitched; however, there may be minor noise present within the stitched portion of the image. Utilizing perspective warping proves effective in diminishing this noise within the stitched image segment.

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